RISC V SOC - MVP Configuration

We recommend 3 viable Minimal viable production configurations

Recommendation

Application	ISA	Pipelining	Privilege	Memory Organization
Simple embedded applications (control/sensing, tiny firmware)	RV32I	1 or 3 in-order	Machine mode	On-chip 8KB-32KB SRAM 4KB Boot ROM
General-purpose (rich bare-metal firmware)	RV32IM	3 or 5 in-order Prefetch Buffer	Machine mode	32-64 KB instruction SRAM + 32-64 KB data SRAM (64-128 KB total) 4KB Boot ROM Prefetch Buffer - 2K
Advanced project (RTOS/Linux)	RV32IMC	5 stage in-order Prefetch Buffer Instruction Cache	Machine mode and Supervisor mode	Atleast 64KB each SRAM memory 4KB Boot ROM Prefetch Buffer - 2K On-chip I Cache - 4-8K

Please note,
MMU will be needed to boot Linux

Recommended peripheral set

Peripheral	# of instance	Usage	Source	Mandatory
32 pin GPIO	1	controlling, configuring, and reading the digital state of the physical pins	https://github.com/pulp-platform/gpio	Yes
UART	1-2	one for debug, one for application communication	https://github.com/pulp-platform/apb_uart	Yes
SPI/QSPI	1	For sensor comms/flash access & Booting vis SPI Flash	https://github.com/pulp-platform/apb_spi_master	Yes
I2C	1	Sensor communication	https://github.com/pulp-platform/apb_i2c/blob/master/apb_i2c.sv	Can be used if needed
I2S	1	orchestrator of digital audio transmission	??	Can be used if needed
PLIC	1	Interrupt Controller supporting 16-32 interrupt sources	https://github.com/pulp-platform/rv_plic/blob/master/rtl/plic_top.sv	Yes
Timer	1	System tick, general-purpose	https://github.com/pulp-platform/apb_timer	Yes
CLIC/CLINT	1	Timer and SW Interrupts	https://github.com/pulp-platform/clic	Yes

Another open source peripheral: https://opencores.org/projects

Recommended Interconnect

For core to peripheral communication, AXI4-Lite:
https://github.com/pulp-platform/axi/blob/master/src/axi_lite_to_apb.sv

Recommended VIPs

https://mbits-mirafra.github.io/projects/

Boot Plans

Boot ROM and/or SPI	**Teams can have a BootROM and load the instruction memory before the CPU reset is de-asserted.** **We also recommend adding a SPI slave in the peripheral list, so that boot code can be fetched from an external SPI memory.** **Need external boot mode selection pin.** **When 1'b0 then Boot code is fetched from internal ROM else boot code is fetched from external SPI memory.** **SPI is a backup option if boot from ROM fails.**	Teams need to plan for the verification of both the paths